Design and tone in the mechanoacoustic piano. Part I. Piano hammers and tonal effects[a)]

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Relations between the design of a piano and the quality of its tone are reviewed and discussed from the point of view of an instrument designer. Topics include the effects on tone by the hammers, the voicing, the soundboard, the case, the string plate, and the design of the scale and strings, including longitudinal string modes, wrapped strings, and inharmonicity. Design limitations relating to materials, performance, or tone quality are noted. The effect of varying a parameter is described or illustrated graphically when feasible. Data comparing pianos of various size and design are shown. Examples are shown of the use of mobility measurements in assessing soundboard and string plate performance. The potential value of modal and finite element analysis in design work is noted. Although the piano is regarded by many people as fully developed, the author believes that there has never been a better time for improvement. Some directions for future work are indicated. © *1996 Acoustical Society of America.*

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INTRODUCTION

The oldest existing piano, completed in 1720 by its Florentine inventor, Bartolomeo Cristofori (1655–1731), is on display at the Metropolitan Museum of Art in New York City.[1] The instrument is 2.29 m long and has 54 keys (C2 to F6) with two strings per key. The Museum has issued good recordings of it by an excellent pianist.[2,3] To listen to these in comparison with almost any good recording of a modern piano is to gain quickly an impression of the possible degree of influence of design on tone quality, for the sound of the old instrument is indeed different from that of pianos today. The 1720 Cristofori has been described in an excellent article by Stewart Pollens, Master Restorer at the Museum.[4] Information from that article has been used here to compare original and modern designs. In order to concentrate here on acoustical factors, certain significant mechanical differences, e.g., in the action/key mechanism, will be mentioned only briefly. These differences are also musically important since much of todays standard repertoire probably would be unplayable on the 1720 instrument because of its simpler, less controllable action.

I. HAMMERS

Hammers define the mechano-acoustic piano and are important elements in determining its timbre. However, although it is absolutely necessary to have ''good'' hammers in order to produce satisfactory piano tone, the hammers merely transmit exciting forces to the strings. A poor piano with even the best hammers will still be judged to have poor tone.

In the earliest pianos there was little or no intentional difference between bass and treble hammers. The hammers now in the 1720 Cristofori (probably not the original ones) have wooden cores with a thin leather covering to contact the strings. They appear to be substantially identical across the scale.[5,6] The hammers were glued onto wooden shanks that pivot near their opposite ends in a way that resembles present practice.[7] To Cristofori, a builder of harpsichords, it may have seemed intuitively correct for all of the hammers to have identical properties, so as to present an equal load to the fingers, no matter what key was played. Eventually it became clear that better tone can be produced with hammers that differ. If the hammers of the 1720 Cristofori were to be installed in a modern instrument, the tone probably would be thin, weak, and twangy, especially in the bass, and generally unsatisfactory by modern standards.

Contemporary bass hammers are much larger than treble hammers. Figure 1 shows the No. 1 (10.1 g) and No. 88 (3.8 g) hammers for a Steinway grand. The hammers have hardwood cores of graduated size. These ''moldings'' are covered with one or two layers of wool felt that increase in thickness from treble to bass. The outer layer for each set of hammers is formed from a strip of felt about 1 m long and 10 cm wide that increases in thickness lengthwise from treble to bass, and that also tapers in thickness widthwise from the center to each outer edge so that any cross section has the approximate shape of an isosceles triangle (Fig. 1). In a press that is long enough to accommodate a complete set of hammers, the felt strip is stretched and compressed around a set (88 plus spares) of wooden moldings[8] that are assembled side-against-side, and is glued to the moldings. After the felt strip and the moldings are removed from the press, individual hammers are sliced from the strip with a special knife. Samples of outer felt, removed from hammers after pressing, have been found to have densities in the range 0.6–0.7 g/cm^3.

Hammer mass is a critical factor for optimum perfor-

[a)]This invited review and tutorial article is being published in three parts. Part II—Piano structure, and Part III—Piano strings and scale design will appear in the near future. Final editing and publication were unintentionally but unduly delayed, for which the Editor-in-Chief apologizes to the author.

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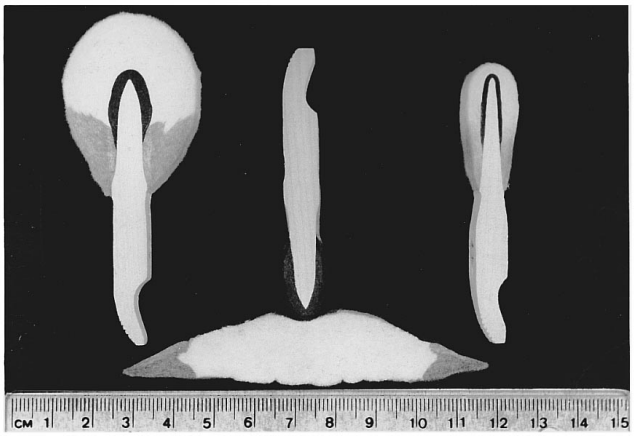


FIG. 1. No. 1 (10.1 g) hammer, and No. 88 (3.8 g) hammer for a Steinway grand. Center: Bass hammer with outer felt (below) removed, showing approximate initial shape of felt cross section.

mance, especially in the treble. Figure 2 gives a representative mass curve for a set of grand piano hammers. Hammer mass (m) typically increases over the 88-note compass by a factor between 2 and 3. The ratio (m/M) of hammer mass to string mass varies more widely. Figure 3 shows (m/M) for a 2.1-m grand piano. The M value for a key is the ''speaking mass,'' the total mass of the speaking lengths of the strings for that key.

Even though the spectrum of a piano tone changes in a complex way as a function of hammer velocity, the output peak sound pressure level varies approximately in direct proportion to hammer velocity, increasing by about 6 dB as the hammer velocity at impact with the strings doubles. This relationship applies for hammer velocities up to at least twice those obtainable in normal use (Fig. 4). Maximum and minimum attainable output levels for a piano of given acoustical efficiency depend on a combination of physical, mechanical, manufacturing, and human factors. Minimum useful output is determined by the design and regulation of the action mechanism and by the skill of the pianist in judging and supplying to the key that minimum energy which will reliably produce a tone (with less input energy the hammer may never reach the strings). In the extreme treble, the author finds that some increase in acoustical output for a given final hammer velocity could be obtained by making those hammers somewhat lighter than shown in Fig. 2, but this would increase the difficulty of hammermaking. The bass fundamental and low-order partials could be increased by making the bass hammers heavier, but then the instrument would become harder to play, and the subjective loudness for a given hammer velocity could be reduced.

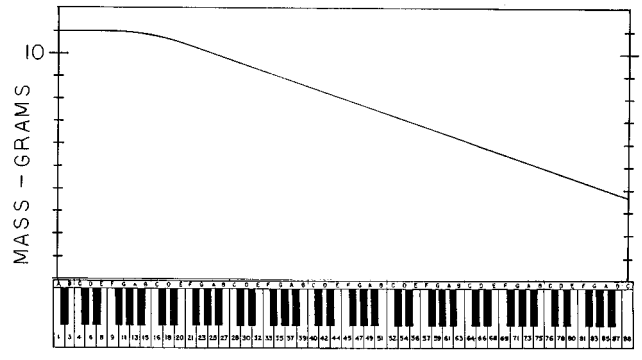


FIG. 2. Representative mass curve for grand piano hammers.

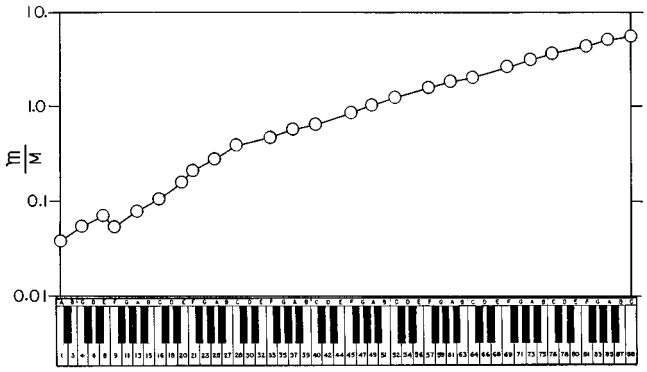


FIG. 3. Ratio (m/M) of hammer mass to string speaking mass for a representative grand piano.

In modern pianos the distance of the hammers from their rest position to the strings ranges between 40–50 mm (from make to make), but in any individual instrument it is regulated to be constant across the entire compass. Likewise, the key travel from rest to fully depressed is adjusted to be constant (about 10 mm) all across the keyboard. In consequence, the customary treble-to-bass increase of hammer mass produces a corresponding graduation of key impedance, making it impossible for all keys to present the same finger load. Thus there is a physical basis for the complaint of some pianists that they cannot get around the keyboard as nimbly with the left hand as with the right. The standard compensating method, used in instruments of the best quality, is to install lead weights permanently in the keys at such positions that the same weight (about 50 g), when placed very gently on the playing surface of each key while the dampers are

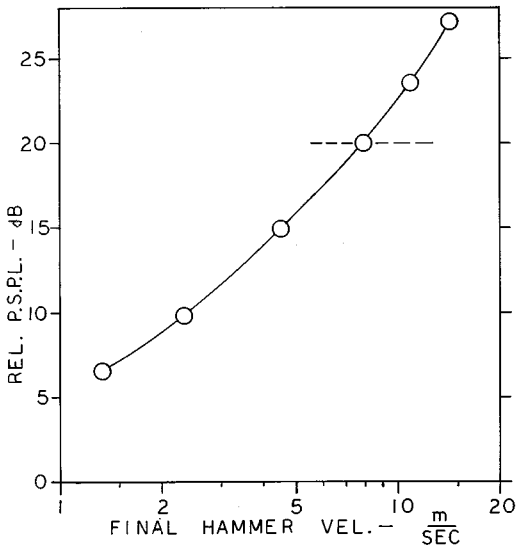


FIG. 4. Relative envelope peak sound pressure level (P.S.P.L.) versus final hammer velocity for grand piano tone C1. Maximum with key playing is shown by dashed horizontal line. Higher hammer velocities were obtained with a special hammer-driving means.

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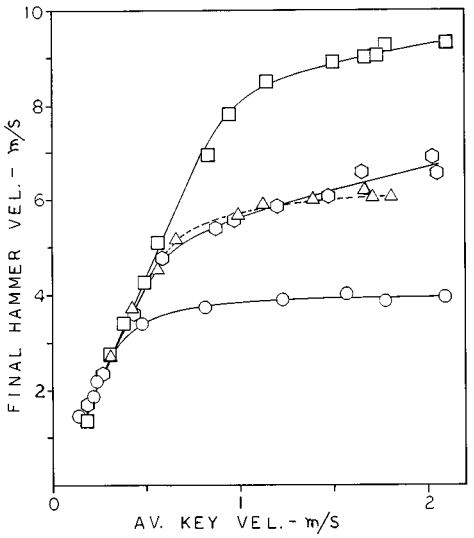


FIG. 5. Final hammer velocity versus average key velocity for two different grand piano actions, showing limiting effect. The circles are for Key 3 of Piano A. The hexagons are for Key 87 of Piano A. The triangles are for Key 3 of Piano B. The squares are for Key 87 of Piano B. Differences between the curves are due to differences in action and key design and hammer mass.

lifted, is required to initiate key motion all across the keyboard. Skilled pianists can easily detect ± 5 g. This equalization of static ''touch weight'' subjectively normalizes key impedance for pianissimo playing, but has little effect at high levels, for which the required key force can be 100 *N* or more, originating mostly from accelerating the hammer to its final velocity. Often there are differences from pianist to pianist in the value of ''touch weight'' that is preferred, and remarks are heard frequently that one action is ''too heavy'' and another ''too light.'' Normal small differences in hammer mass or in the mechanical advantage of the action- and -key set can be noticed. The amount of friction in the action/key mechanism is another significant variable. The equalization of touch with fixed compensating weights has the disadvantage that the touch cannot be adjusted by the pianist, and only with difficulty by the piano technician. Schemes for making the touch weight adjustable by the pianist have been devised but have not come into common use.[9]

The input impedance of a piano key and the maximum attainable-hammer velocity are limited by the masses and compliances of the various parts. Figure 5 shows curves of final hammer velocity versus average key velocity for Key Nos. 3 and 87 of two different grand pianos, indicating the degree to which the design of the action/key/hammer mechanism can affect the maximum output of a piano. Typically, hammer velocities of 4 to 5 m/s are readily attained, but beyond this a limiting region is reached within which a further increase in key velocity will not increase the hammer velocity and acoustical output in the same proportion as at lower dynamic levels. As observed by Askenfelt and Jansson,[10–12] it is possible at high dynamic levels for a key to reach the bottom of its stroke before the hammer has touched the strings. This indicates that energy is being stored temporarily in the compliance of the key. Reference 12 contains an excellent study of the modern grand action mechanism.

In grand pianos the mechanical system of the hammer head on its wooden shank normally has a major vibrational mode in the region of 200–300 Hz when installed in the piano. This mode, excited when the hammer strikes the strings, is not normally audible in the lower half of the piano seale, but can be heard as a pitched noise component or ''knock'' in the treble register. (The pitch of this knock can be identified by playing individual treble keys while heavily damping the strings to prevent them from sounding.) Random variations in stiffness from shank to shank can be obvious and annoying when playing successive treble notes one at a time, so the shanks should preferably be similar. The frequency of the shank mode can also influence noticeably the treble tone quality of the instrument, lower system frequencies generally being more favorable to the tone. In recognition of this, treble hammer shanks are often made thinner and more flexible than those in the lower part of the scale. The mechanism of this apparent relation between shank system frequency and tone quality, an effect evidently separate from the knock itself, is not yet clear but may involve interaction between the motion of the hammer head due to the shank resonance and the motion of the strings against the hammer head due to transverse waves propagating in both directions on the string. Suzuki may have found the key to this in his study[13] of the effect of shank vibration on the force-time pattern applied to a rigid string. He found that a hammer with a stiffer shank has a broader force-time pattern than with a more flexible shank. The results of Askenfelt and Jansson[14] showing an unexpected increase in hammer–string contact time at C6, may be related to this.

The motion of the hammer as it contacts the strings is extremely complex. In addition to Refs. 9–14, several other articles have recently appeared.[15–24] In addition to new results, Hall's articles contain excellent reviews of older, precomputer era work on this problem. Several of these papers deal with the nonlinear compliance of the hammer and related effects. The effective hardness of the hammer increases with increasing hammer velocity due to inherent properties of the felt and to the gradient in felt hardness from the outside to the inside of the hammer. Hammer–string contact time decreases with increasing hammer velocity. This is responsible for the long-recognized changes in tonal output spectrum that occur with increased dynamic level. As the tone becomes louder it also becomes brighter. Further studies of the relation between nonlinearity and the changes in timbre with dynamic range would be of particular interest.

The heavier a piano hammer, the longer it will stay in contact with the strings. There is a critical range, normally between G4 and G6, where the contact time of the hammer against the string equals the transit time for transverse vibration along the string.[25,26] Below this range the hammer leaves the string before the arrival of the first major reflection from the bridge. Above this range the hammer normally will still be in contact. Due to the above noted nonlinearity, the location of the critical region may shift in frequency with hammer velocity. If the hammer is in contact at the time of the first reflection, losses can occur that cause an undesirable quality of tone. Clearly, energy lost in this manner will be unavailable for acoustical purposes. Figures 6 and 7 show the attenuation of acoustical output caused by a grossly over-

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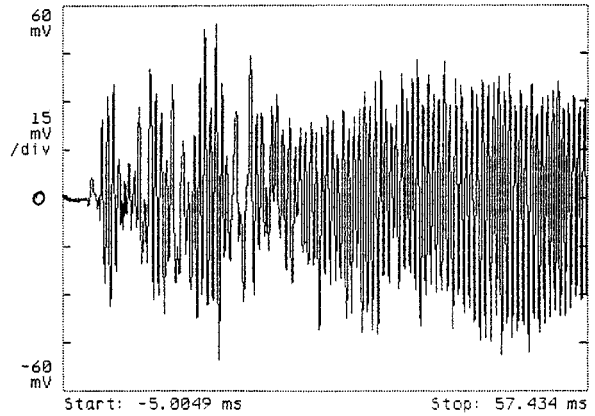


FIG. 6. Onset time trace for G6 with hammer of normal (6.4 g) mass.

weight G6 hammer. Figure 6 shows the onset time trace obtained with a hammer of normal mass (6.4 g). Figure 7 shows the onset time trace after a 4.8-g mass was attached to the wooden molding of this same hammer. A mechanical playing device maintained in both cases the same average key velocity so the input energy actually increased with the heavier hammer; but the output at the fundamental frequency decreased by nearly 9 dB, and the tone took on a disagreeable ''choked'' quality. The effect of over-weight hammers is particularly critical in the treble region, where hammers only a gram or so too heavy can cause a noticeable deterioration in tone quality.

It is inevitable, with real hammer and string parameters, that throughout much of the high treble the hammers will be in contact with the strings when transverse vibrations return from the bridge. In order to produce a satisfactory tone within this critical region, it is important to have an optimum hammer striking position along the strings, to have the hammer strike all of the unison strings equally, and to maintain the mass of the hammer within an acceptable range. These criteria are interdependent. Piano technicians may sometimes have to experiment a bit with such things as hammer fit to the strings, felt texture, striking point, etc., in order to produce the best possible result. For example, when the offensive sound of an overweight treble hammer is heard, it can often be eliminated by moving the striking point of the hammer closer to the nearer end of the string so as to reduce the hammer–string contact time. Hall[27] has predicted and found discontinuities of hammer–string contact and other phenomena during the contact period which, it seems intuitively, might explain most of the observed results. Above the critical region, the control of tone quality is more straightforward, but the striking point of the hammer along the strings becomes increasingly sensitive for maximum acoustical output. In grand pianos the hammer–string contact time and the tone quality are radically modified by use of the una corda pedal, which shifts all hammers a few millimeters laterally toward the treble so that (except for the low bass strings) one string remains un-struck and behaves as a coupled passive resonator. This also affects radically the rate of tonal decay. The elegant articles of Weinreich[28–30] show in some detail the effects of the *una corda*, which relate to the coupling between the multiple strings of a given key via the soundboard and bridge. In upright pianos the left-hand pedal merely moves all the hammers closer to the strings without shifting them laterally. This reduces the hammer velocity for a given key velocity, which affects the tone as noted below, but not (in theory) the decay rate.

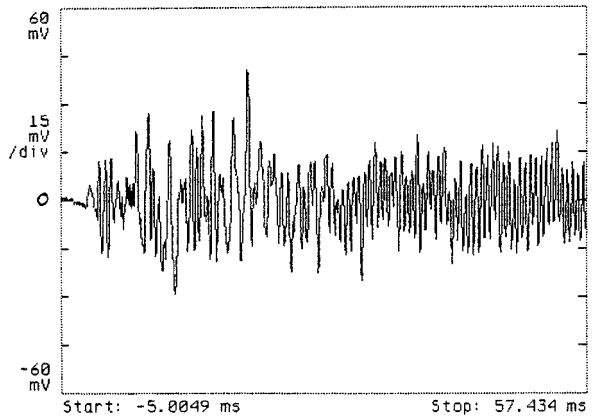


FIG. 7. Onset time trace for G6 with 4.8 g added to normal hammer.

Piano tone spectra vary as a function of dynamic level. The levels of the high-frequency partials change more than for low-frequency partials with a given change in hammer velocity.[31] In other words, the change in timbre from pianissimo to forte entails not only ''turning up the volume'' but also ''turning up the treble tone control.'' This effect contributes importantly to making the piano musically exciting to hear. The amount of the change depends on the properties of the hammer. Figure 8(a) and (b) illustrates a normal amount of onset spectrum change at G5 for tones at ff and pp levels. At the fundamental frequency (784 Hz) the level difference between ''very loud'' and ''very soft'' measured 24.7 dB. (With a properly regulated grand action a dynamic range of about 30 dB is achievable.) For the ff tone [Fig. 8(a)] the level of the 2nd partial was about 12 dB below that of the fundamental, and the level of partials 3–7 ranged between 30–38 dB lower. For the pp tone [Fig. 8(b)] the 2nd partial was about 35 dB below the fundamental and partial 3 was more than 50 dB below.

The FFT analyzer used for this measurement (Hewlett–Packard model 35660A) could compute and display ''total harmonic distortion.'' The ff tone was said to have 25.9% THD, while the pp tone showed only 2.1%.

What was being measured was of course not really ''distortion,'' and its presence certainly did not make the tone sound bad. There is, however, a type of distortion peculiar to pianos that is related to hammer properties and it does sound bad. This distortion, when present, sounds something like peak clipping in an audio amplifier, and it usually occurs at high playing levels, often in very bright pianos or pianos that have been played a lot without attention to the hammers. The effect has occasionally been referred to by pianists as ''playing through the piano'' because the objectionable quality normally has an onset threshold related to hammer velocity. Players who hear the trouble can reduce or eliminate the symptoms by keeping the dynamic level below the threshold. This unexplained distortion may be caused by an excessively

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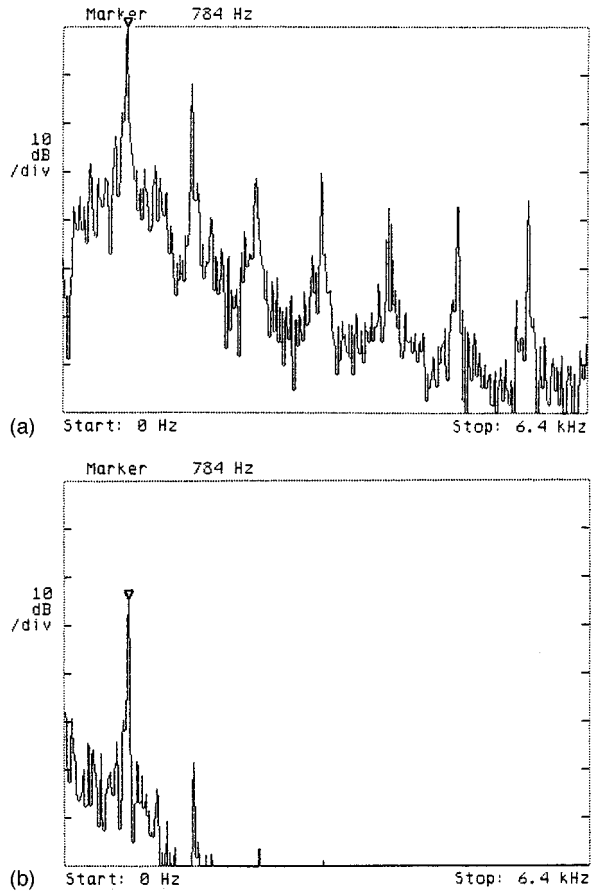


FIG. 8. (a) Onset spectrum for G5 played ''ff'' with normal felt hammer. (b) Onset spectrum for G5 played ''pp'' with normal felt hammer.

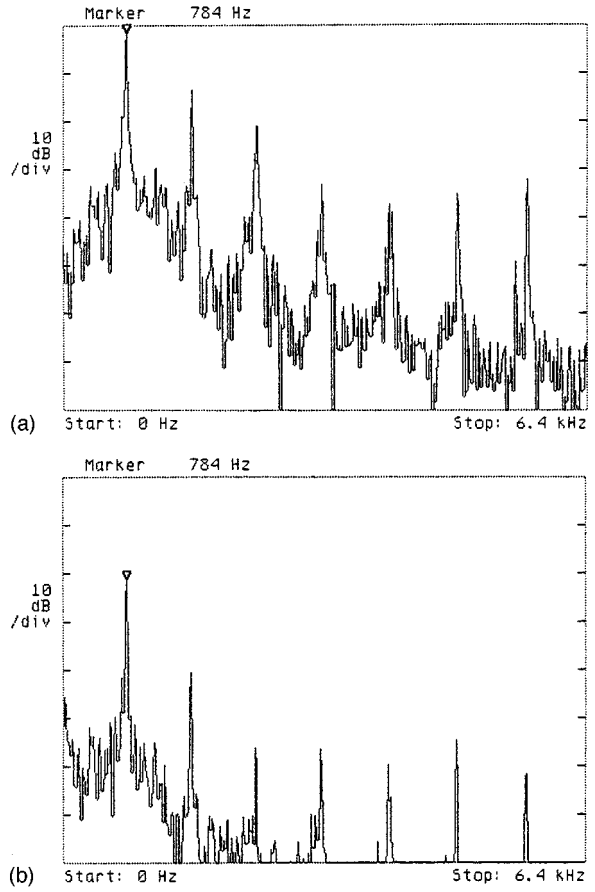


FIG. 9. (a) Onset spectrum for G5 played ''ff'' with wooden hammer. (b) Onset spectrum for G5 played ''pp'' with wooden hammer.

large gradient of hammer hardness, possibly in combination with hammer–string contact phenomena, and perhaps aggravated by permanent flattening of the hammer from use.

The hardness of piano hammer felt directly affects the overall quality of the tone. For the best tone, each hammer must have its hardness within a certain range, and the hardness must have a gradient such that the string-contacting surface is softer than the inner material. The existence of a hardness gradient is believed by the author to be chiefly responsible for the spectral changes that occur with changes of dynamic level. The effect of a too-small hardness gradient is illustrated by Fig. 9(a) and (b) which shows spectra for a ff tone and a pp tone, in comparison to Fig. 8(a) and 8(b), for which the hammer had a normal hardness gradient. The hammer for Fig. 9(a) and (b) (4.95 g) had its hardness gradient greatly reduced by removal of enough felt so that the strings were struck by the wooden molding. The tones produced sounded very bright and somewhat harpsichord-like, but the ff and pp tones had similar timbre. The fundamental of the pp tone [Fig. 9(b)] was about 20 dB lower in level than that of the ff tone. If 20 dB were added to each partial of the pp tone, its spectrum clearly would more nearly resemble that of the ff tone than would be the case if a similar adjustment were made to the spectrum of Fig. 8(b) before comparison with Fig. 8(a).

A ''Shore A'' durometer having a small contacting ''foot'' [32] has been used on the sides of hammers to obtain an indication of hardness and gradient. Figure 10 shows a hammer marked with five points for such a measurement, the points shown being identified nominally as ''0%'', ''25%'', ''50%'', ''75%,'' and ''100%'' of the width of the white outer felt. Figure 11 shows the hardness, measured by this method, of three different hammers of similar size and weight. The vertical axis represents the upper portion of the ''Shore A'' hardness scale. (On the ''Shore A'' scale a flat piece of glass measures 100.) The horizontal axis gives the location of the measured points as shown by Fig. 10. Gradient is indicated by the slope of the hardness curve near the outside of the hammer felt. These three hammers were installed in turn at G5 of the same piano and played at the same level to test the tone. The softest hammer (lowermost curve, Fig. 11) produced a pleasant tone that was perhaps a bit too soft (dark). The medium hammer (middle curve) produced a significantly brighter and louder tone that was judged to be within the optimum quality range. With the hardest hammer (top curve) a still brighter tone was produced that was judged to be somewhat harsh and contained objectionable extraneous noise components. Figures 12(a)–(c) are spectrograms produced with these hammers. The acoustical output at the fundamental frequency changed very little, but the amplitude of the the upper partials decreased as hammer hardness decreased. The output at partial 7 was about 23 dB less for the softest hammer than for the hardest.

The optimum hardness for piano hammers varies widely

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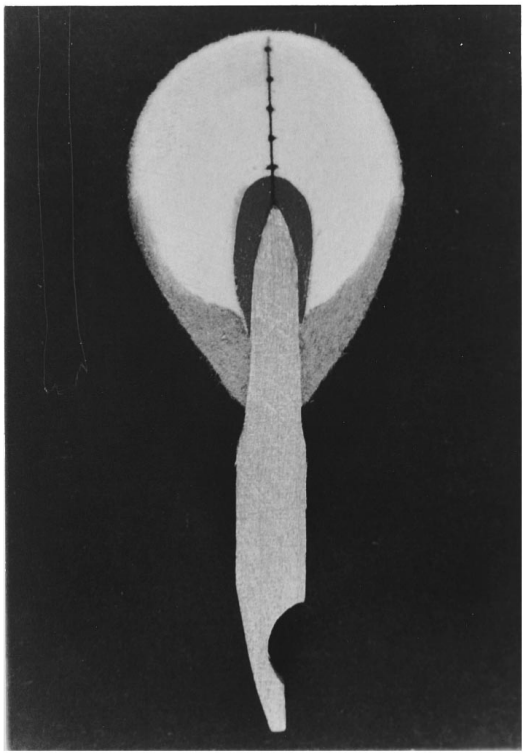


FIG. 10. Location on hammer of points measured with durometer.

with key number. The extreme treble hammers need to be much harder than the middle or bass hammers in order to produce tones of the same loudness. The need to make the hammers harder in the treble usually begins, probably not by coincidence, in or near the critical region in which the hammer contact time against the string and the propagation time for transverse waves along the string become equal. Figure 13 is a generalized curve that shows approximately how the relative Shore ''A'' hardness of hammers at the 75% point changes across the scale, in order to give tones judged by pianists to have ap-

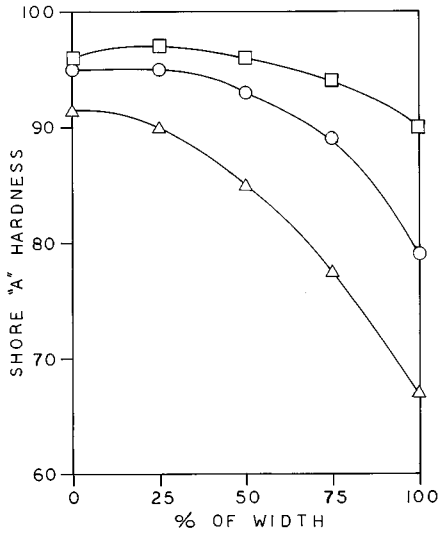


FIG. 11. Measured Shore ''A'' hardness of three different G5 hammers of similar weight, size, and shape.

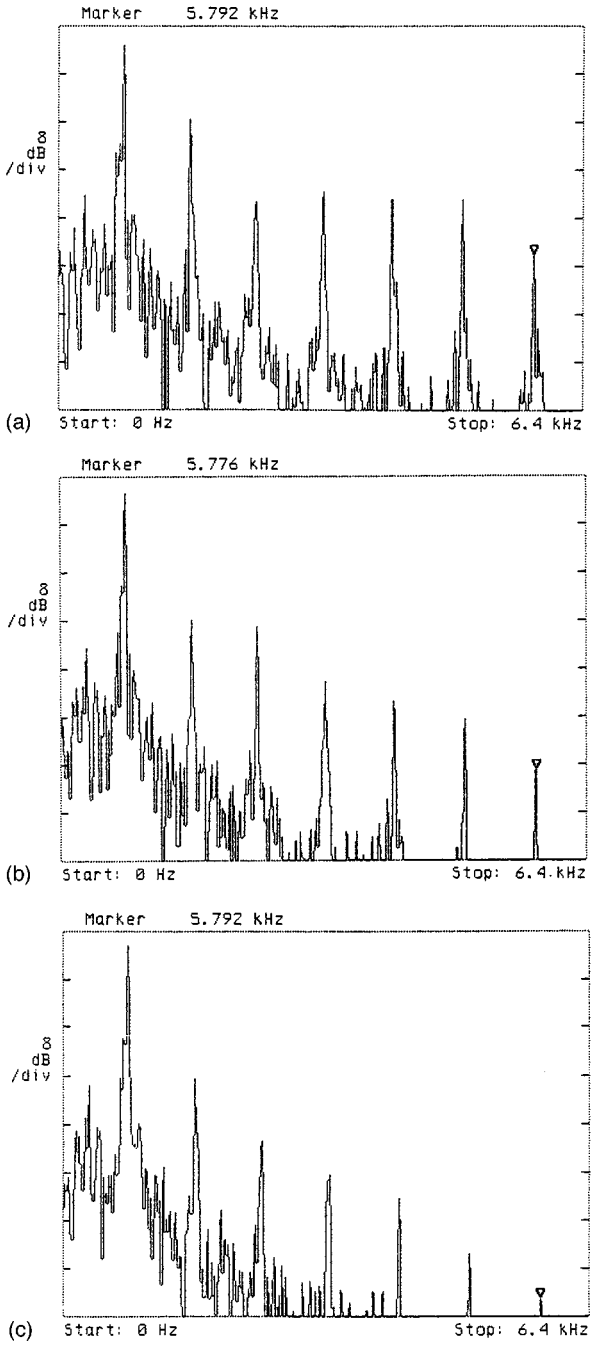


FIG. 12. (a) Onset spectrum of G5 produced with hardest hammer (hardness curve identified by square data points in Fig. 11). (b) Onset spectrum of G5 produced with medium hammer (hardness curve identified by circular data points in Fig. 11). (c) Onset spectrum of G5 produced with softest hammer (hardness curve identified by triangular data points in Fig. 11).

proximately equal loudness for equal key effort, a condition considered desirable because it makes artful playing a little easier. The overall tonal brightness (and hammer hardness) that will be considered optimum often varies widely from listener to listener, and depends somewhat on the acoustical environment and on the characteristics of the individual instrument as well as on the music. For these and other reasons, concert pianists prefer to select an instrument for a particular performance from a group of several of varying tonal character.

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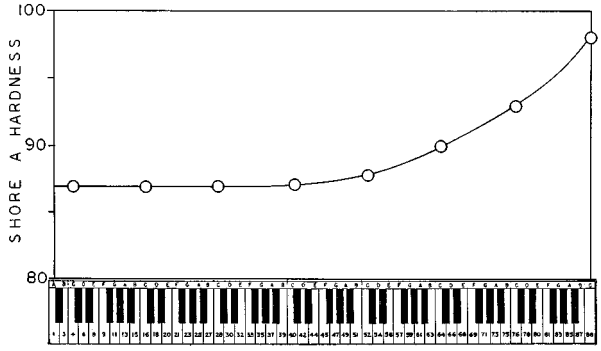


FIG. 13. Approximate Shore ''A'' hammer hardness required to give tones of equal loudness for equal key effort.

II. VOICING HAMMERS

Wool felt[33–36] is advantageous as a hammer covering material because hammers made with it can be ''voiced.'' Voicing, also termed tone regulating or toning, is a process of manipulating the hardness, the gradient, and other properties of piano hammers in order to produce the ''best'' tone and to maintain a timbre that is agreeably consistent from key to key across the keyboard.[37,38] Although not a design factor, voicing deserves mention here because it can be one of the most important influences on the final ''personality'' of a piano. In extreme cases, voicing can transform an instrument no pianist wants into an instrument universally liked. Voicing, normally done only after a piano has been completely assembled, mechanically regulated, and perfectly tuned, is sometimes omitted in less costly instruments. Hammers can be made harder so as to produce a brighter (and usually also louder) tone with more energy at higher frequencies by ''filing'' away with sandpaper some of the softer outer felt. If the underlying felt is not hard enough, a hardening agent (such as a solution of nitrocellulose lacquer or acrylic plastic in a quick-drying solvent) can be applied with care in selected correct places on each hammer. If a hammer is initially too hard, the felt can be softened by being ''needled'' (pierced with needles of sewing size) in carefully selected areas, near to but seldom directly on the part that contacts the strings. This produces relaxation and consequent softening of the felt. The changes in the output spectrum due to voicing a mid-treble hammer are essentially the same as those resulting from differences in hammer hardness as illustrated by Figs. 12. Moreover, the effects of voicing can produce changes in measured hardness similar to those in Fig. 11. Voicing is reversible if not carried too far or repeated too often. The effect of voicing at F3 is indicated by Fig. 14(a)–(c). Figure 14(a) shows the acoustical spectrum at onset given by a hammer that produced acceptable tone. Next the hammer was needled extensively, yielding the spectrum of Fig. 14(b), which shows a negligible change in the fundamental but a large reduction in the level of the high partials. The tone was then judged to be too soft. Next, a solution was applied to increase the hardness of the hammer beyond the original value. This gave the spectrum of Fig. 14(c), producing a very loud and bright tone that was still acceptable in quality. The use of hardening solutions, although not a new practice, has been controversial among piano technicians,

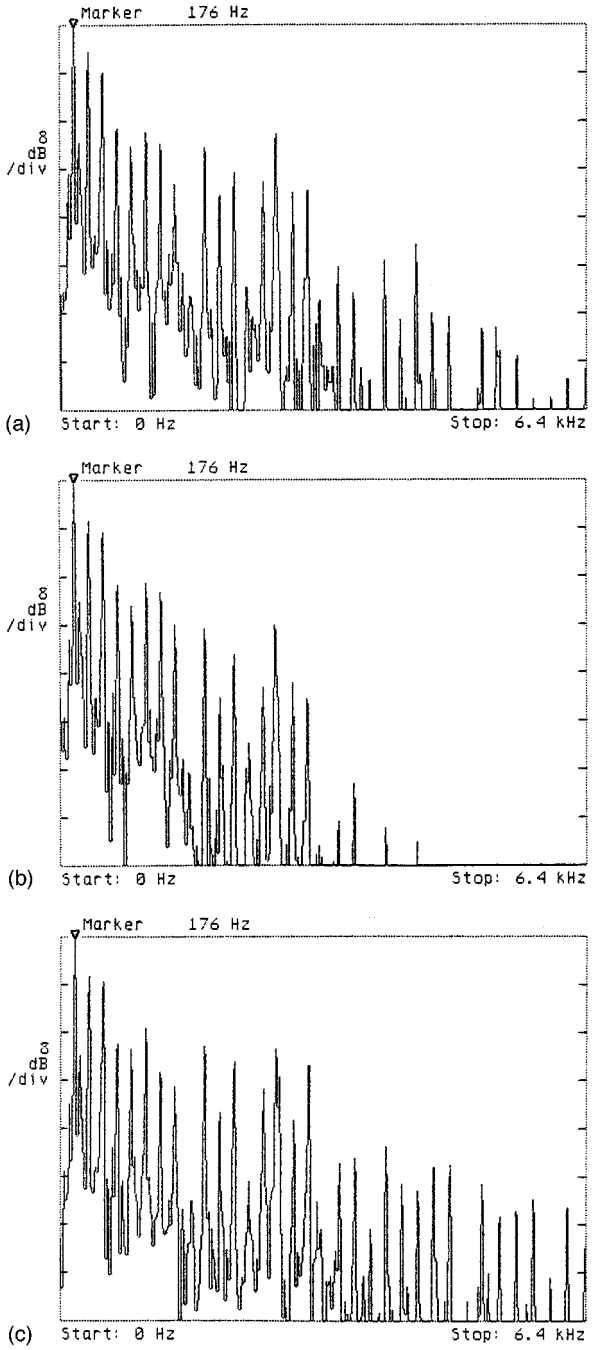


FIG. 14. (a) Onset spectrum of F3 produced with a normal hammer giving acceptable tone. (b) Onset spectrum of F3 produced with the hammer after extensive needling, giving excessively soft tone. Note that level of fundamental is essentially unchanged. (c) Onset spectrum of F3 produced with needled hammer of (b) after treatment with a hardening solution, giving very loud and very bright tone of acceptable quality. Note that level of fundamental is essentially unchanged.

there being some who feel it is essential to good tone, others who feel that it is injurious to hammers and that only sandpaper or in some cases a hot iron or a steam iron should be used. [Note: In Fig. 14(b) a prominent partial is visible at about 2.6 kHz. This partial which was affected much less than the others by voicing, is not a transverse partial but instead is the first longitudinal mode of the string. A discussion of the longitudinal modes of piano strings appears later.]

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In the lower part of the piano scale, voicing affects primarily the upper partials, causing relatively little change in the amplitude of partials below 1–2 kHz. Voicing the extreme treble hammers affects all of the partials, the higher the partial the more affected. The tonal effect of adjusting hammer hardness is similar to the effect of adjusting the treble tone control of an audio amplifier, with the tone control having a corner frequency around 1–2 kHz. Harder hammers slope the treble response up, softer hammers slope it downward. Voicing is more than the mechanical and straightforward process of adjusting the spectra of piano tones. Voicing is really the art of knowing what to do to make the best-sounding instrument and being able to do it. As in all arts, practitioners vary in skill and talent. Any experienced voicer, given a decent piano and good hammers, can make it musically acceptable. An exceptional few can go beyond this to produce an instrument that is able to sound extraordinary. The rest is up to the pianist.

The musical potential of a piano can be limited by hammer properties in several ways, but the following are the most basic: If the felt in the central core of the hammer is too soft by durometer test, then no amount of filing can give the result wanted. The choices are either to accept a dull sounding piano or to harden the hammers by chemical or other means. Occasionally hammers may be encountered that are too hard to be needled effectively. Either the needles cannot be inserted into the felt with a reasonable amount of force or, if they can be inserted, no relaxation response occurs. The ability of hammers to respond to voicing is influenced by their design, by the materials used, and very strongly by the manufacturing techniques employed.

There has existed much lore having to do with the relation between the properties of the felt used to make hammers and the tonal results obtainable. The best hammers have been said to require felt that has been fulled from a critically proportioned combination of wools from different kinds of sheep, including usually at least some found only in places at a great distance. Felt to be used for piano hammers is routinely tested for such properties as physical dimensions, hardness, and density, but other factors evidently are also important, for hammers made in exactly the same way with felt from different sources often exhibit significantly different properties in use, even though the felt has met the same specifications. Tests of various alternate hammer materials have so far not disclosed any that are superior to wool felt. Reasons for rejection have included lack of a gradient, production of extraneous noises, and inability to be voiced. The piano industry awaits a better hammer.

III. WHERE THE HAMMER SHOULD STRIKE THE STRINGS

Early piano makers did not fully appreciate the effect of varying the striking ratio d/L, where L is the speaking length of the string and d is the distance along the string from the hammer striking point to the nearer speaking length termination. Figure 15 shows d/L plotted versus key position for the 1720 Cristofori and for a representative contemporary grand. The values for the old piano vary over a wide range to no apparent purpose, while those of the modern instrument follow a generally smooth curve within a more restricted range. It was not until the late 18th century that a consensus began regarding d/L.[39] The consensus, reported by many authors, was that for best tone the hammers should strike the strings between 1/7 and 1/9 of their speaking length.[40–51] That conclusion may seem puzzling to anyone now who works with pianos because the 1/7 to 1/9 rule is inconsistent with good treble tone, and is routinely violated in present instruments. Possible explanations for this discrepancy may be that modern instruments have a more extended treble compass and heavier hammers than early pianos, and/or that some authors may have accepted earlier conclusions without verification.

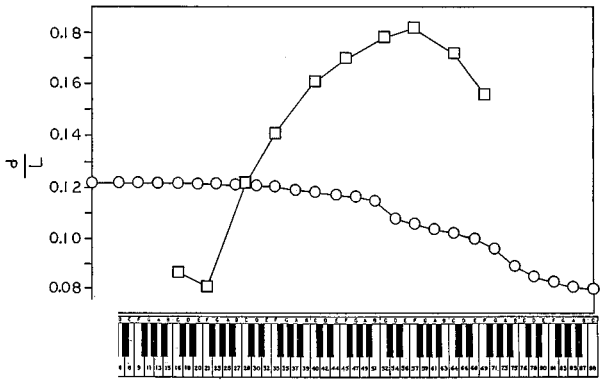


FIG. 15. Hammer striking ratio (d/L) for a contemporary grand (circles) and for the 1720 Cristofori (squares).

The theoretical effect of varying d/L has long been known. Reference 21 contains a detailed discussion of this subject and an extensive bibliography. In theory, an ideal stretched string struck at a certain point will not vibrate at frequencies for which vibrational nodes exist at the striking point. For actual piano strings a large but not infinite attenuation of the predicted partials does occur. Figures 16 and 17 give measured results that the author obtained by varying d/L for a string 1.37 m long, designed for and tuned to A0 (27.5 Hz). The string was mounted on a heavy steel test frame and struck by a standard piano hammer of typical weight for which the velocity at the instant of striking was held constant. The vibration of the string was sensed by a pickup located on the terminating mass at one end of the speaking length. Figure 16 shows the results for partials one

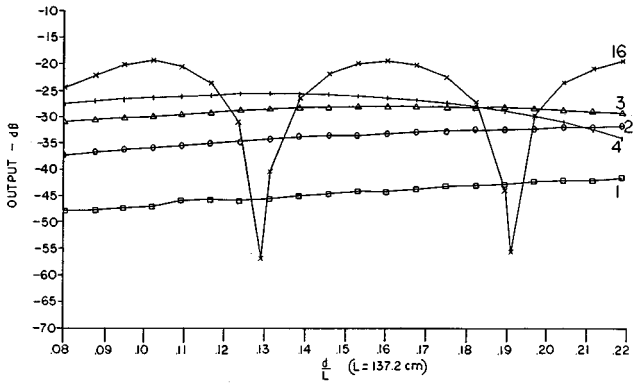


FIG. 16. Output level (d/L) of partials 1–4 and of partial 16 for an A0 (Key 1) string of 1.37 m long, mounted on a test frame.

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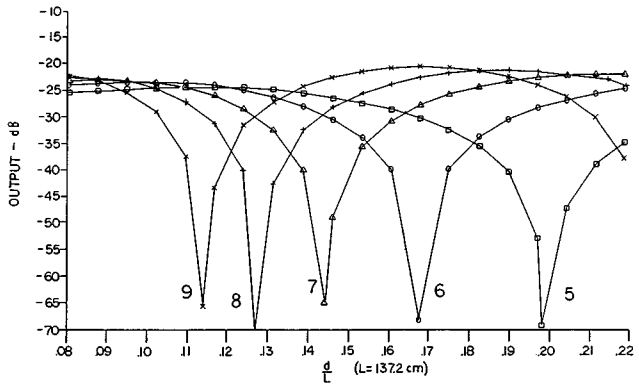


FIG. 17. Output level versus (d/L) of partials 5–9 for an A0 (Key 1) string 1.37 m long, mounted on a test frame.

through four and for partial 16. Figure 17 shows partials five through nine. Figure 18 shows spectrograms at $d/L=0.019$ and $d/L=0.143$ for another A0 string 2.10 m long. The data show minima typically 45 to 50 dB below the maximum output of a partial. Timbre changes could easily be heard in the amplified output of the sensor, and also in the acoustical output of a piano used in a similar test in which (d) was varied. Based on analogy between a vibrating string and an electrical transmission line, Kock[52] concluded that partials of a real string cannot be completely eliminated by varying d/L because of the presence of finite losses. Hall[53] came to the same conclusion. Because of increased losses due to the bridge and soundboard, and possibly also in part because of the large number of other resonators present in the system,

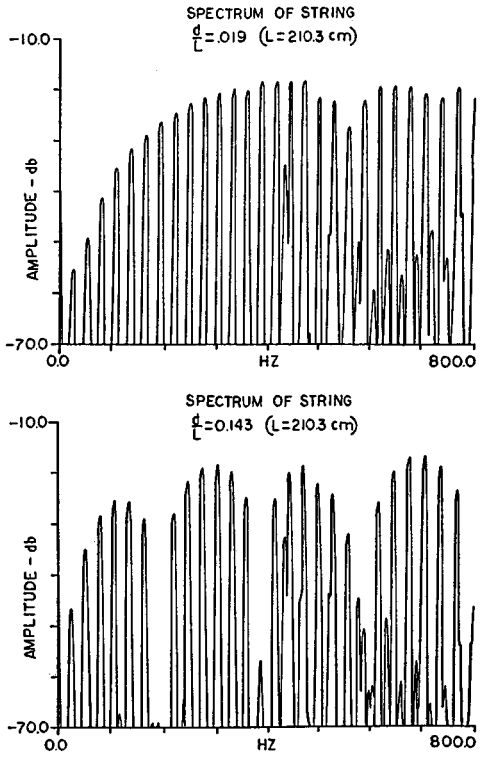


FIG. 18. Spectrograms for $(d/L)=0.019$ and also for $(d/L)=0.143$ for an A0 (Key 1) string of 2.10 m long mounted on a test frame.

TABLE I. d/L values that produced a ''B-flat Maj. scale'' from a B-flat (Key #2) string installed in a 1.32-m vertical piano. String data: $L=1.365$ m; core=1.24 mm; Cu underwrap=0.518 mm; Cu outer wrap=1.47 mm; speaking mass=208 g; Calc. $T=954.8$ N.

Tone	d (cm)	d/L
do	22.2	0.163
re	20.2	0.146
mi	18.1	0.133
fa	17.1	0.125
sol	16.0	0.117
la	14.9	0.109
si	13.7	0.100
do	12.6	0.0923

minima in the acoustical output for strings in a real piano are usually less deep than those measured on a test frame. Random examples can be seen in Fig. 14, where partial 9 appears about 28 dB down. In these examples d/L was neither chosen nor adjusted for maximum attenuation, so the true minima may be deeper than shown.

There seems no doubt that musically inclined listeners, if allowed to compare tones initiated at different d/L values, will be persuaded that certain values are tonally preferable to others. I have demonstrated that in certain cases it is possible, by varying only the striking point, to produce from the same string a sequence of tones resembling a major scale, as the formant due to d/L varies.[54] Approximate values of d/L to produce the ''tones'' of this scale are given in Table I for the particular string tested. Recently, d/L values as large as about 1/6 have been used in certain pianos and claimed to be tonally advantageous in the low bass, and it has been indicated that tonally optimum values of d/L vary with the amount of inharmonicity of the string.[55] Some writers believed that the 7th and 9th partials cause a dissonant or unpleasant tone and ought to be eliminated.[56] In none of the testing I have done has there ever been an indication that the quality of a real piano tone can be improved in the general case by minimizing any specific partials or partials.

The criteria for deciding where the hammer should hit the strings are different for the treble than for the bass. In the low bass, timbre varies significantly with d/L, while acoustical output changes rather gradually. (Note the relative flatness, within the normal range for d/L, of the curves for partials 1–6 in Figs. 16 and 17.) The aim is therefore to choose d/L to give the most pleasing tone, as well as sufficient output. In the extreme treble, large changes in acoustical output occur as the striking point of the hammer is moved, but timbre changes are relatively small, so the object is simply to maximize the output. That a pronounced maximum exists can be verified by moving the treble end of the action and keys of a grand piano slightly in and out of the piano case as Key 88 is being played. This is done routinely where grands of high quality are made because it is necessary for best results to locate the optimum point empirically for each individual instrument. The optimum d/L value at Key 88 varies with hammer weight and shape and also with string size and contact conditions, but normally occurs for d/L between 1/12 and 1/17. For reasons given later, the design parameters for the extreme treble of contemporary pi-

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anos vary only slightly with make and model. However, I was recently surprised to note a Key 88 hammer that was positioned at a second output maximum located farther from the end of the string than the primary maximum normally used, the reason apparently being that a mechanical interference precluded use of the primary maximum, which would have given more output. (George[57] observed that multiple output maxima can occur for large values of m/M.)

In the mid-treble, the best d/L value is often a compromise between maximum first partial energy and the most graceful tone. The optimum value is affected by hammer weight. Reducing the striking distance (d) in the mid-treble generally will make the tone sound ''thinner'' because less first partial (fundamental) energy will be present. Increasing the striking distance may make the tone sound ''fatter,'' but too large a (d) will produce an unclear, ''muddy'' quality due to excessive hammer dwell time, and will decrease the apparent loudness of the tone. In most well-designed contemporary pianos, the value of d/L remains relatively constant throughout the lower half of the compass and decreases gradually and smoothly from around mid-scale to extreme treble. In the bass of the largest grand pianos, d/L tends to approach a non-acoustical limit because the larger the value of (d), the longer the keys have to be. For a typical 2.74-m concert grand in which the No. 1 string is just over 2 m long and d/L is about 1/8, the length of the shortest possible No. 1 key is about 0.6 m, which presses practical limits both for the pine wood normally used to make keys and for the performance of the action. In upright pianos the value of d/L is unrelated to the length of the keys.

[1] The Metropolitan Museum of Art, New York, NY 10028, The Crosby Brown Collection of Musical Instruments, 1889. Piano Forte (89.4.1219).
[2] The Metropolitan Museum of Art, New York, NY 10028, LP Record MMA L1803: THE CRISTOFORI PIANOFORTE—1720 (1978).
[3] The Metropolitan Museum of Art, New York, NY 10028, LP Record MMA L1802: WINDS AND STRINGS (1978).
[4] S. Pollens, J. Am. Musical Instrum. Soc. **10**, 32–68 (1984) (America Musical Instrument Society, University of South Dakota, Box 194, Vermillion, SD 57069).
[5] Reference 4, pp. 40–43.
[6] The Metropolitan Museum of Art, New York, NY 10028, The Crosby Brown Collection of Musical Instruments, 1889, Piano Forte (89.4.1219), Photograph: Negative No. 129116 B, ''Detail: Action.''
[7] Reference 4, pp. 41–42.
[8] G. A. Briggs, *Pianos, Pianists and Sonics* (Wharfdale Wireless Works, Bradford, York, U.K., 1951), p. 59 (photo of clamped set of moldings, felt strip, and lower caul).
[9] For example, H. A. Conklin, Jr. and P. W. van Nuis, U.S. Patent No. 4,381,691, ''Touch Force Adjustment Means for a Piano'' (3 May 1983).
[10] A. Askenfelt and E. Jansson, ''From touch to string vibrations—The initial course of the piano tone,'' J. Acoust. Soc. Am. Suppl. 1 **81**, S61 (1987).
[11] A. Askenfelt and E. Jansson, ''From touch to string vibrations,'' in *Five Lectures on the Acoustics of the Piano*, edited by A. Askenfelt (Royal Swedish Academy of Music, Stockholm, 1990).
[12] A. Askenfelt and E. Jansson, ''From touch to string vibrations. I: Timing in the grand piano action,'' J. Acoust. Soc. Am. **88**, 52–62 (1990); see also **90**, 2383–2393 (1991) and **93**, 1–16 (1993).
[13] H. Suzuki, ''Vibration analysis of a hammer-shank system,'' J. Acoust. Soc. Am. Suppl. 1 **81**, S83 (1987).
[14] Reference 12, Fig. 9, p. 61.
[15] T. Yanagisawa and K. Nakamkura, ''Dynamic compression characteristics of piano felt,'' J. Acoust. Soc. Jpn. **40**, 725–729 (1984) (in Japanese).
[16] H. Suzuki, ''Modeling of a piano hammer,'' J. Acoust. Soc. Am. Suppl. 1 **78**, S33 (1985).
[17] D. E. Hall, ''Piano string excitation in the case of small hammer mass,'' J. Acoust. Soc. Am. **79**, 141–147 (1986).
[18] D. E. Hall, ''Piano string excitation II: General solution for a hard narrow hammer,'' J. Acoust. Soc. Am. **81**, 535–546 (1987).
[19] D. E. Hall, ''Piano string excitation III: General solution for a soft narrow hammer,'' J. Acoust. Soc. Am. **81**, 547–555 (1987).
[20] H. Suzuki, ''Model analysis of a hammer-string interaction,'' J. Acoust. Soc. Am. **82**, 1145–1151 (1987).
[21] D. E. Hall and P. Clark, ''Piano string excitation IV: The question of the missing modes,'' J. Acoust. Soc. Am. **82**, 1913–1918 (1987).
[22] X. Boutillon, ''Model for piano hammers: Experimental determination and digital simulation,'' J. Acoust. Soc. Am. **83**, 746–754 (1988).
[23] D. E. Hall and A. Askenfelt, ''Piano string excitation V: Spectra for real hammers and strings,'' J. Acoust. Soc. Am. **83**, 1627–1638 (1988).
[24] D. E. Hall, ''The hammer and the string,'' in *Five Lectures on the Acoustics of the Piano*, edited by A. Askenfelt (Royal Swedish Academy of Music, Stockholm, 1990).
[25] J. P. Quitter, ''Research and Development on the Piano,'' IRE Trans. Audio **AU-6** (September–October), 96–103 (1958) (see pp. 101–102).
[26] Reference 23.
[27] References 17–19.
[28] G. Weinreich, ''Coupled piano strings,'' J. Acoust. Soc. Am. **62**, 1474–1484 (1977).
[29] G. Weinreich, ''The Coupled Motion of Piano Strings,'' Sci. Am. **240** (1), 118–127 (January 1979).
[30] G. Weinreich, ''The coupled motion of piano strings,'' in *Five Lectures on the Acoustics of the Piano*, edited by A. Askenfelt (Royal Swedish Academy of Music, Stockholm, 1990).
[31] For another example see Ref. 22, Fig. 4, p. 748.
[32] Standard Rubber Hardness Tester, JIS Cat. #250-181 (diameter of contacting surface=1 cm) manufactured by: Shimadzu Seisakusho Ltd., 1 Nishinokyko-Kuwabaracho, Nakagyo-ku, Kyoto 604, Japan.
[33] *Wool Handbook*, edited by W. Von Bergen (Wiley, New York, 1963), 3rd ed., Vol. 1 (Vol. 1 covers raw materials prior to manufacturing, including physical and chemical properties of wool, breeds of sheep, grading of wool, etc.).
[34] *Wool Handbook*, edited by W. Von Bergen (Wiley, New York, 1969), 3rd ed., Vol. 2, Pt. 1 (Vol. 2, Part 1 covers the manufacture of woolen yarn and fabrics. It includes material on sorting and cleaning procedures for raw wool but is not otherwise relevant to felt making).
[35] *Wool Handbook*, edited by W. Von Bergen (Wiley, New York, 1970), 3rd ed., Vol. 2, Pt. 2 (Vol. 2, Part 2 covers dyeing, finishing, carpets, felts, and testing. Sections of interest in relation to felt include Chap. 15, pp. 1075–1119, *re*: the manufacture of felts, both nonwoven and woven, and perhaps Chap. 16, pp. 1121–1214, ''Physical and Chemical Testing'').
[36] N. Hyde, ''Wool, Fabric of History,'' Natl. Geographic **173**(5), 552–591 (1988).
[37] F. R. Deitz, *Das Intonieren Von Flugeln (Grand Voicing)* (version in German, English, French, Swedish, and Italian, with 16 photos) (Verlag Erwin Bochinsky GmbH, Munchener Strasse 45, W-6000 Frankfurt/Main 1, Federal Republic of Germany, 1968).
[38] A. Kuerti, ''What Pianists Should Know About Pianos,'' Clavier (May-June), 12–19 (1973) (see pp. 17–19).
[39] R. E. M. Harding, *The Piano-Forte—Its History Traced to The Great Exposition of 1851* (Da Capo, New York, 1973), reprint of 1933 Ed., pp. 64–66.
[40] H. L. F. Helmholtz, *Sensations of Tone as a Physiological Basis for the Theory of Music* (Dover, New York, 1954, Orig. Pub. Braunschweig, 1863), p. 77.
[41] E. Brinsmead, *The History of the Pianoforte* (Singing Tree Press, Detroit, MI, 1969, Orig. Pub. Novello, Ewer and Co., London, 1879), p. 47.
[42] S. Hansing, *Das Pianoforte in Seinen Akustischen Anlagen* (C. Laffert, Breslau, 1888), p. 92.
[43] W. B. White, *Theory and Practice of Pianoforte Building* (Edward Lyman Bill, New York, no date, foreword dated 1909), p. 34.
[44] S. Wolfenden, *A Treatise on the Art of Pianoforte Construction* (Unwin Bros., Ltd., Old Working, Surrey, U.K., 1975, Orig. Pub. 1916, reissued with Supplement, 1927), pp. 50–51.
[45] O. Ortman, *The Physical Basis of Piano Touch and Tone* (Dutton, New York, 1925), p. 96.
[46] A. Vant, *Piano Scale Making* (A.V. Vant, New York, 1927), p. 30.
[47] A. Wood, *The Physics of Music* (Methuen, London, 1944), p. 94.
[48] Reference 8, p. 37.

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[49] W. V. McFerrin, *The Piano—Its Acoustics* (Tuners Supply Co., Boston, 1972), p. 84.
[50] S. Marcuse, *Musical Instruments, a Comprehensive Dictionary* (Norton, New York, 1975), p. 405.
[51] E. M. Good, *Giraffes, Black Dragons, and Other Pianos* (Stanford U.P., Stanford, CA, 1982), p. 9.
[52] W. E. Kock, ''The vibrating string considered as an electrical transmission line,'' J. Acous. Soc. Am. **8**, 227–233 (1937).
[53] Ref. 21, p. 1914, ''Conclusion 6,'' p. 1916.
[54] H. A. Conklin, Jr., Lecture-demo. given at Piano Acoustics Conference, Steinway and Sons, New York, 20 May 1989.
[55] H. A. Conklin, Jr., U.S. Patent No. 4, 674, 386, ''Augmented Bass Hammer Striking Distance for Pianos'' (23 June 1987).
[56] See, for example, W. B. White, *Piano Tuning and Allied Arts* (Tuners Supply Co., Boston, MA, 1946), 5th ed., p. 45.
[57] W. H. George, ''The Partial Tones of the Struck String—Part I,'' Philos. Mag. **50**(6), 491–499 (1925).

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